Credit Risk Analytics

Loan Default Analytics for Lending Decisioning

1. Project Overview

This project analyzes a lending institution's loan portfolio using transactional and applicant data from 31,522 loan applications (after cleaning), covering borrower demographics, loan characteristics, and repayment outcomes. The goal is to identify which borrower and loan attributes are actually associated with default, evaluate whether current interest-rate pricing is risk-adjusted, and produce specific, actionable underwriting recommendations for a credit risk team.

2. Dataset Summary

- **Raw rows:** 32,581
- **Cleaned rows:** 31,522 (after cleaning - 902 rows removed for impossible age/employment values, 157 duplicates removed)
- **Columns:** 12 (raw) + 7 engineered features

Key features

- Borrower demographics: person_age, person_income, person_home_ownership, person_emp_length
- Loan details: loan_intent, loan_grade, loan_amnt, loan_int_rate, loan_percent_income
- Credit bureau history: cb_person_default_on_file, cb_person_cred_hist_length
- Outcome: loan_status (0 = no default, 1 = default)

Missing data

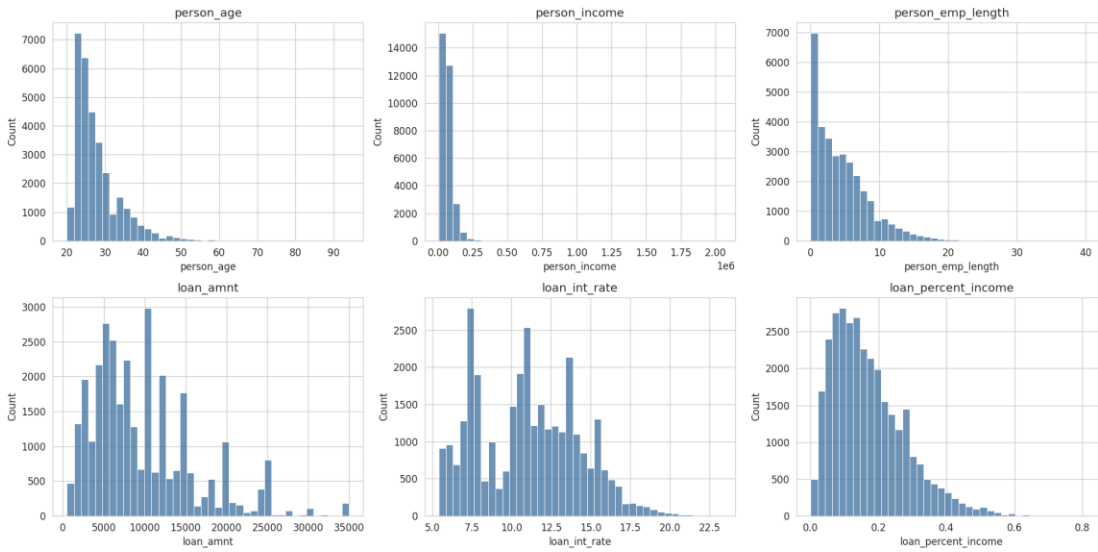
- loan_int_rate: 3,116 rows missing (9.56%) - missingness varied by loan grade, so imputed with the median within each grade rather than a single global value
- person_emp_length: 895 rows missing (2.75%) - imputed with the overall median

3. Exploratory Data Analysis using Python

Data preparation and analysis was carried out in Jupyter notebooks using pandas, NumPy, Matplotlib, and Seaborn.

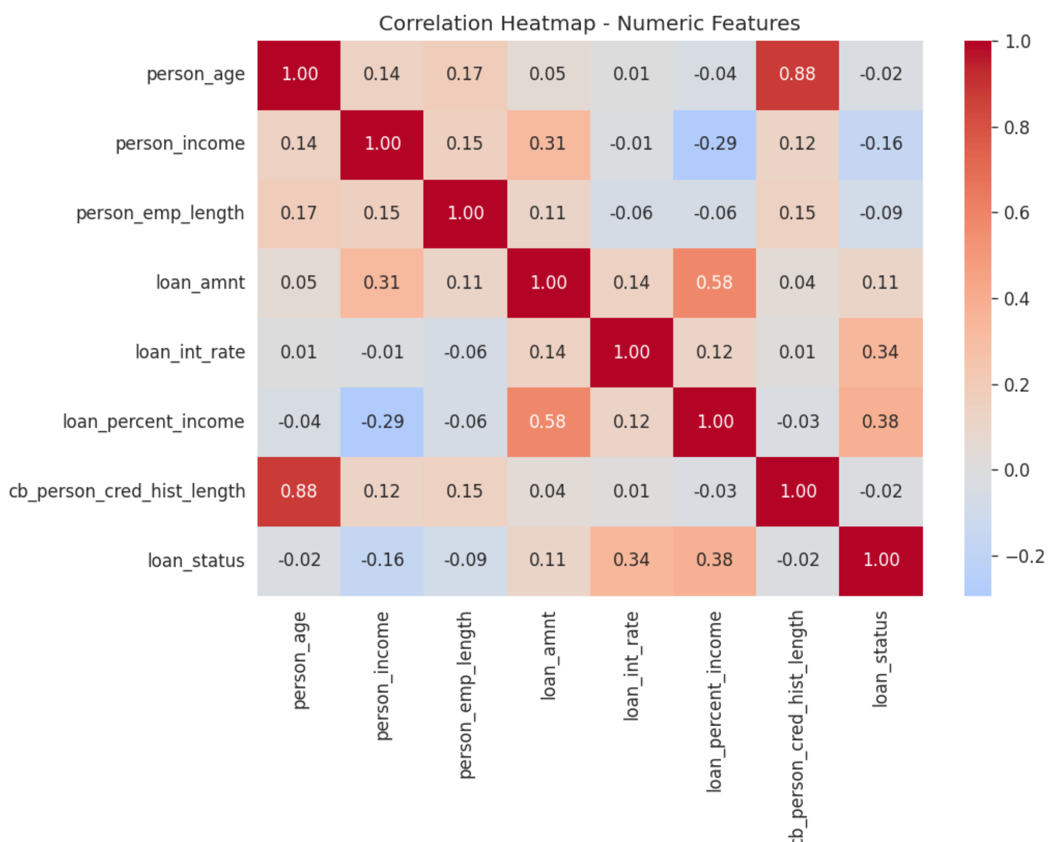
- **Data Loading & Initial Exploration:** Loaded the raw CSV with pandas and profiled structure using .info() and .describe().
- **Missing Data Handling:** Quantified missing values as a percentage of total rows before deciding on an imputation strategy, rather than applying a blanket fill.
- **Outlier Detection:** Identified biologically impossible records (e.g., employment length exceeding age minus 14) and removed them; flagged extreme incomes using the IQR method but retained them, since high earners are plausible, valid applicants.
- **Feature Engineering:** Built risk_tier, dti_bucket, is_renter_or_other, credit_history_bucket, high_risk_intent, income_bracket, and a composite risk_score (0-4), each threshold justified by a pattern confirmed in EDA rather than chosen arbitrarily.
- **Score Validation:** Validated the composite risk_score against actual default rate before recommending its use - confirmed a near-monotonic climb from 5.0% (score 0) to 83.9% (score 4).
- **Database Integration:** Loaded the cleaned, feature-engineered DataFrame into PostgreSQL via SQLAlchemy for SQL-based analysis.

Numeric feature distributions



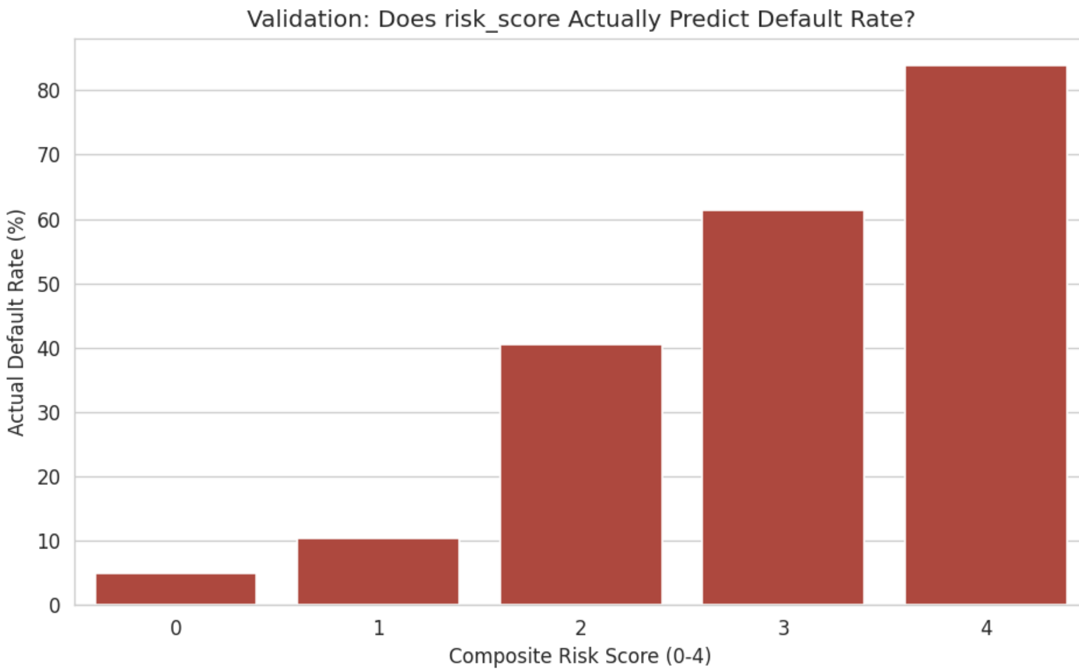
Distributions of person_age, person_income, person_emp_length, loan_amnt, loan_int_rate, and loan_percent_income after cleaning.

Correlation heatmap



Loan-to-income ratio and interest rate show the strongest correlation with default status among numeric features; credit history length is near zero.

Composite risk score validation



risk_score	Default rate	Loans
0	5.0%	8,720
1	10.4%	12,178
2	40.5%	7,500
3	61.4%	2,453
4	83.9%	671

A simple, transparent 4-factor score separates risk 16x from lowest to highest tier - performing well without requiring a black-box model.

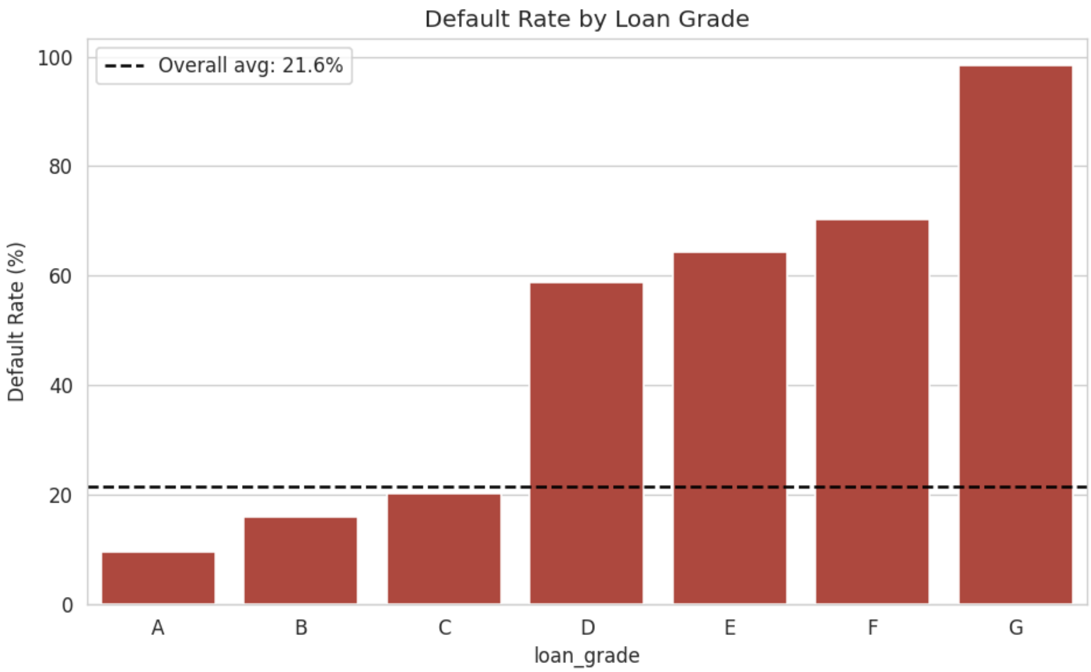
4. Data Analysis using SQL (Business Questions)

Structured analysis was performed in PostgreSQL against a single feature-engineered loans table, using joins-equivalent aggregation, CTEs, window functions, and native statistical functions.

1. **Default Rate by Grade & Home Ownership –** Default rate by loan grade and home ownership.

Grade	Mortgage	Other	Own	Rent	Total
A	4.04%	11.54%	6.55%	16.94%	9.56%
B	7.88%	15.15%	3.59%	23.58%	15.97%
C	14.05%	35.29%	5.92%	25.99%	20.31%
D	45.59%	55.00%	5.95%	73.27%	58.78%
E	47.48%	75.00%	55.56%	75.18%	64.25%
F	63.54%	100.00%	46.15%	77.60%	70.34%
G	100.00%	-	100.00%	96.43%	98.44%
Total	12.50%	31.13%	6.94%	31.12%	21.59%

Even within grade A, renters default at 16.94% vs. 4.04% for mortgage holders - a 4x gap the grade alone doesn't capture.



Default rate climbs from 9.56% (Grade A) to 98.44% (Grade G), with the sharpest jump occurring between Grade C and Grade D.

2. **Pricing vs. Risk by Loan Intent –** Compared average interest rate against actual default rate by loan purpose.

Loan intent	Avg interest rate	Default rate
Debt consolidation	11.06%	28.45%
Medical	11.07%	26.60%
Home improvement	11.25%	25.61%
Personal	11.01%	19.51%
Education	11.00%	16.99%
Venture	10.98%	14.70%

Interest rates are nearly flat (~11%) across all loan purposes despite default rates ranging from 14.70% to 28.45% - pricing is not risk-adjusted by loan intent.

3. **Prior Default History, Controlling for Grade –** Compared default rate for customers with prior default on file, controlling for loan grade (window function, AVG() OVER PARTITION BY).

Without controlling for grade, prior default appeared to be a strong signal (37.6% vs. 18.1% default rate). However, once grade is held constant, the gap nearly disappears (e.g., within grade D: 59.8% with prior default vs. 57.7% without) - a Simpson's Paradox pattern indicating loan grade already absorbs most of the credit-history signal.

4. **Portfolio Exposure vs. Risk Ranking –** Ranked loan grades by dollar exposure and by default rate to find concentration mismatches (RANK() window function).

Grade	Total exposure	Default rate	Exposure rank	Risk rank
A	$88,650,550	9.56%	2	7
B	$101,927,100	15.97%	1	6
C	$58,493,075	20.31%	3	5
D	$38,650,050	58.78%	4	4
E	$12,308,225	64.25%	5	3
F	$3,491,875	70.34%	6	2
G	$1,100,525	98.44%	7	1

Grade B carries the most dollar exposure ($101.9M) but only medium risk - most of the portfolio's capital sits in comparatively safer grades.

5. **Default Rate by Income Bracket –** Ranked income brackets by default rate with a running cumulative share of loan volume.

Income bracket	Default rate
Under $30K	46.54%
$30K-$60K	25.87%
$60K-$100K	13.27%
$100K+	8.99%

The two lowest income brackets alone account for approximately 53.6% of the entire loan book by volume.

6. **Top-Risk Loan Shortlist –** Ranked individual loans by a combined risk_score + loan_amnt priority metric (NTILE window function) to build an underwriting review shortlist.

Identified the highest-priority loans for manual underwriting review - overwhelmingly risk_score = 4, concentrated in grade D, providing a concrete, prioritized list rather than a random sample for review.

7. **Default Rate by DTI Bucket –** Confirmed the debt-to-income bands chosen during feature engineering against actual default outcomes.

DTI bucket	Default rate
Low (<10%)	11.52%
Moderate (10-20%)	14.44%
High (20-35%)	28.74%
Very High (35%+)	71.09%

Default rate jumps from 28.74% to 71.09% once loan-to-income crosses 35% - the sharpest single cliff found in the data, and a defensible cutoff for a hard approval rule.

8. **High-Risk Segment Filtering –** Used a HAVING clause to isolate loan intent x home ownership segments (minimum sample size 100) with default rate above the 21.59% portfolio average.

Seven segments exceeded the portfolio average, concentrated almost entirely in RENT-housing borrowers across medical, personal, venture, education, and debt-consolidation loan purposes.

9. **Statistical Correlation –** Computed Pearson correlation with default status natively in SQL using CORR().

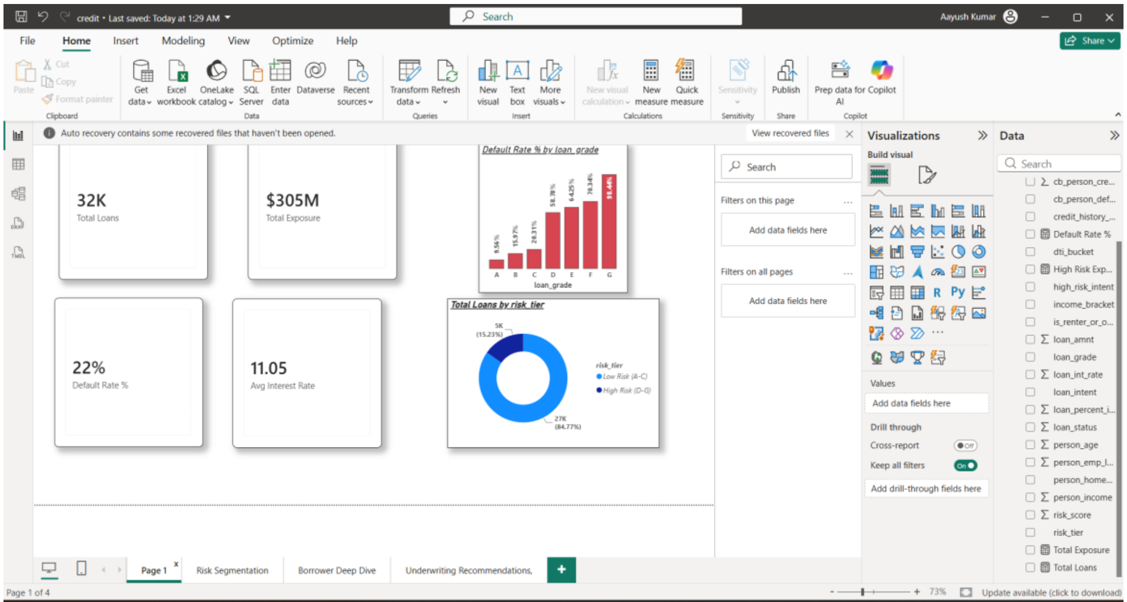
Variable	Correlation with default
Loan-to-income ratio (DTI)	0.3801
Interest rate	0.3383
Person income	-0.1649
Credit history length	-0.0178

Credit history length is essentially uncorrelated with default, confirming the pattern found independently in the Python EDA correlation heatmap.

10. **BI-Ready View –** Created vw_risk_summary as a stable, clean data source for Power BI to connect to.

5. Dashboard in Power BI

An interactive four-page dashboard was built in Power BI, connected directly to the PostgreSQL loans table, to make these findings explorable by a non-technical stakeholder.



Page 1 - Portfolio Overview: KPI cards, default rate by loan grade, and portfolio split by risk tier.

- **Page 1 - Portfolio Overview:** KPI cards (Total Loans, Total Exposure, Default Rate %, Avg Interest Rate), default rate by loan grade, and portfolio split by risk tier.
- **Page 2 - Risk Segmentation:** Default rate matrix (grade × home ownership), default rate by loan intent, default rate by DTI bucket, with interactive slicers for grade, home ownership, and loan intent.
- **Page 3 - Borrower Deep Dive:** Income vs. loan amount scatter (colored by default status), default rate by income bracket, and default rate by credit history bucket.
- **Page 4 - Underwriting Recommendations:** Total vs. high-risk exposure KPI cards ($304.62M vs. $56M, 18.4%), a ranked table of the highest-risk individual loans, a risk-score slicer, and a written recommendation summary.

6. Business Recommendations

- **Tighten approval criteria for Grade D+ applicants –** Default rate nearly triples from Grade C (20.31%) to Grade D (58.78%). $56M (18.4%) of the $304.62M portfolio currently sits in Grade D-G.
- **Reprice loans by purpose –** Interest rates are flat (~11%) across loan purposes despite a 2x gap in default rate between the riskiest and safest purposes - reprice by intent, not just by grade.
- **Apply a hard DTI cutoff at 35% –** Default rate jumps from 28.74% to 71.09% once loan-to-income exceeds 35% - the sharpest single risk cliff identified in the analysis.
- **Deprioritize credit history length as an underwriting factor –** Correlation with default is effectively zero (r = -0.0178); loan grade and DTI are far stronger, evidence-backed signals and should carry more underwriting weight.

Full analysis available in notebooks/01_data_cleaning.ipynb through 04_load_to_postgres.ipynb, sql/analysis_queries.sql, and powerbi/credit_risk_dashboard.pbix.